

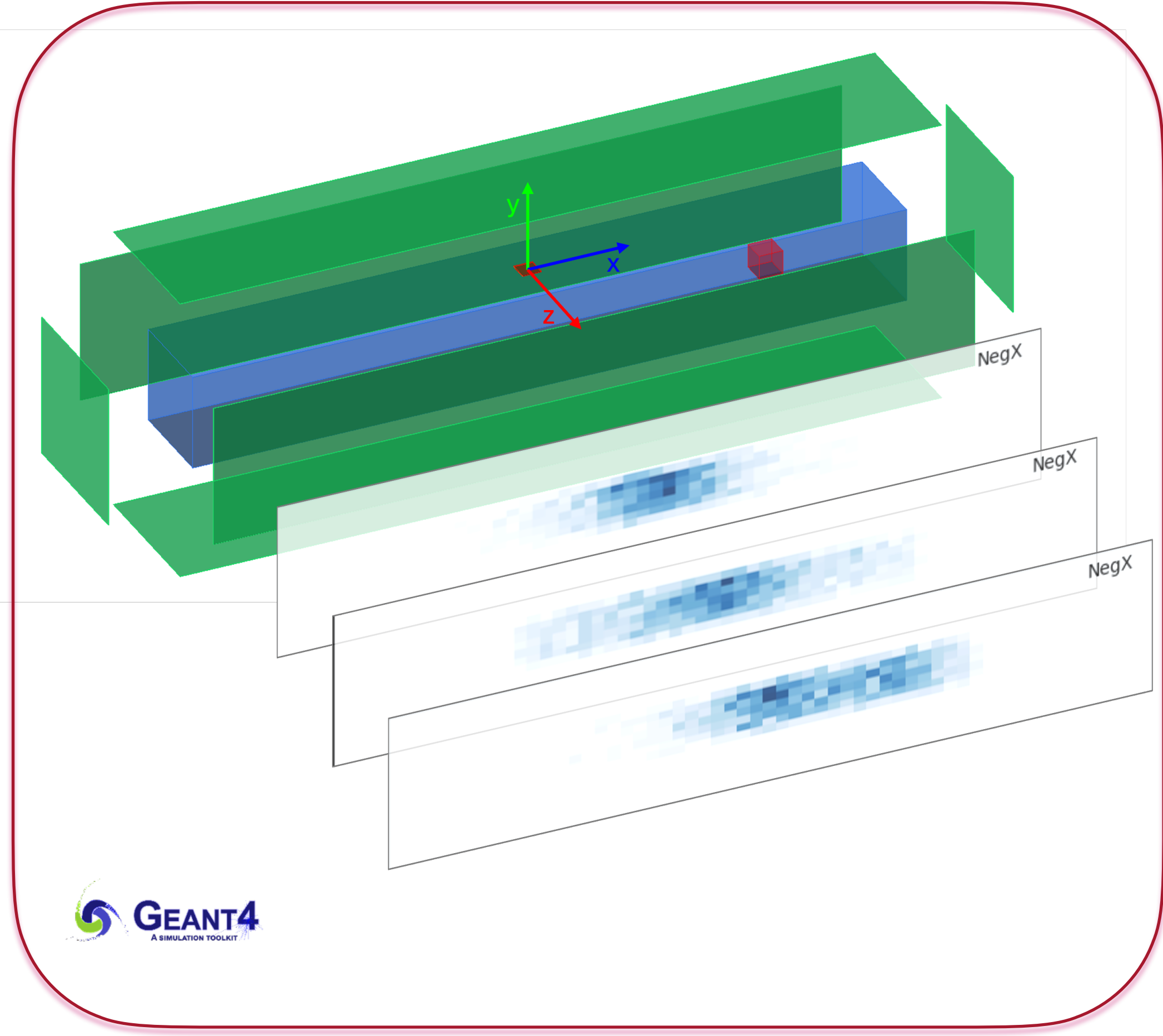
Gamma detection for source localisation in targeted alpha therapy

Abstract

Targeted alpha therapies deliver highly localised dose, but treatment success depends on where the radionuclide source remains after administration. We simulated an AlphaGlue-like geometry with six photon-detector panels and compared retained, broadly diffused, and localised displaced source states across ten repetitions of 10,000 primary decays. Total detector response changed only weakly, from 1.250 +/- 0.008 to 1.206 +/- 0.009 hits per decay, while baseline-subtracted spatial profiles showed clear source-dependent redistribution signatures.

Geometry and display

Gamma entries on six detector panels are used to image source position. The geometry shows the AlphaGlue source, water phantom, detector panels, and example display projections for retained (a), diffuse(b), and localised (c) source states.



Simulation Model

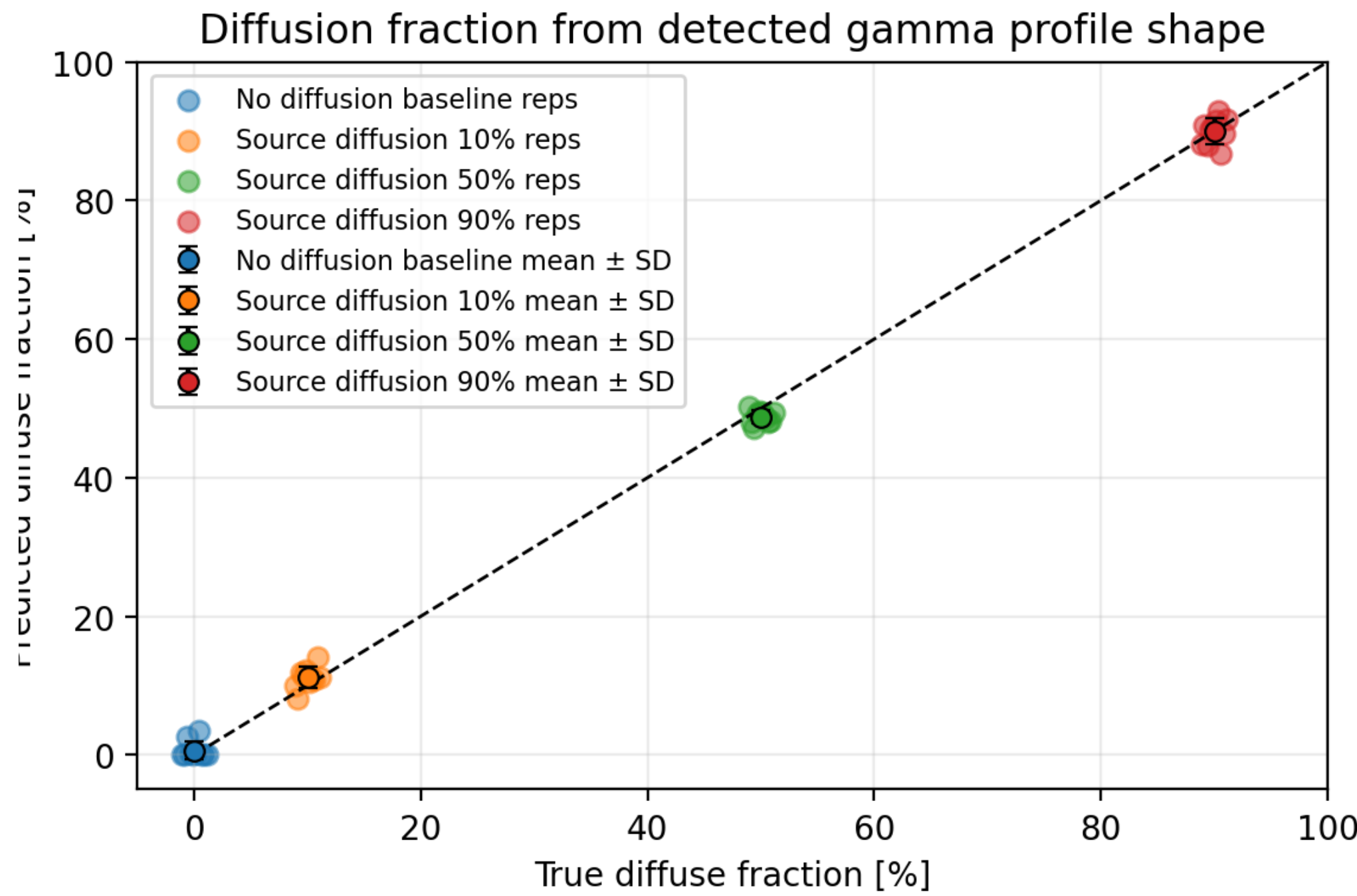
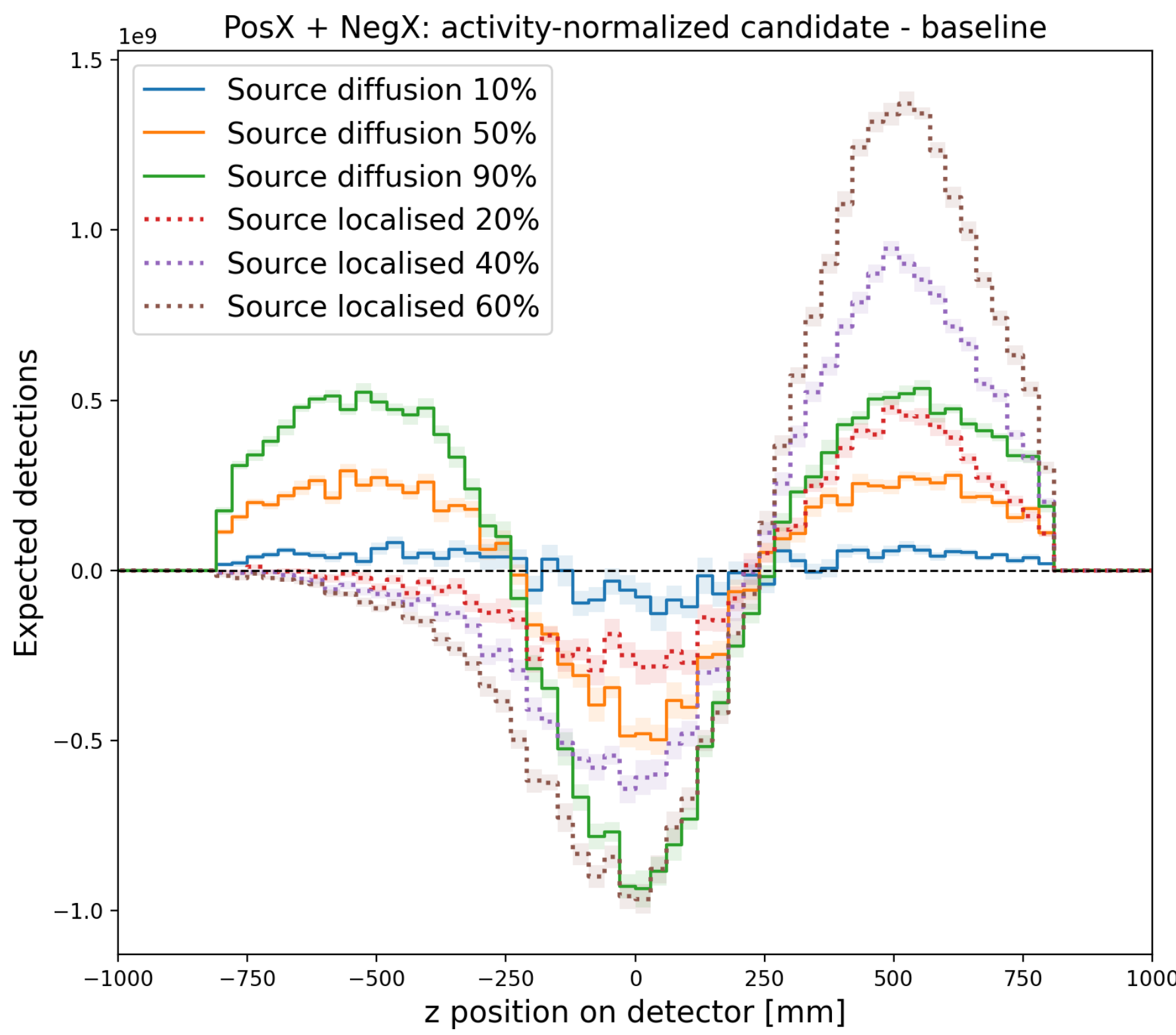
A Geant4 model [4] was used to simulate treatment delivery and gamma detection. The geometry included a water phantom, six photon-detector panels, and a thin AlphaGlue source on the phantom surface. Simulated source states represented retained activity, broad diffusion, and localised displacement, with varying activities.

Source States

CASE	SOURCE STATE
Baseline	100% retained in glue
Diff10/ 50 / 90	10 / 50 / 90% broadly diffused
Loc20/ 40/ 60	20 / 40 / 60% localised displacement

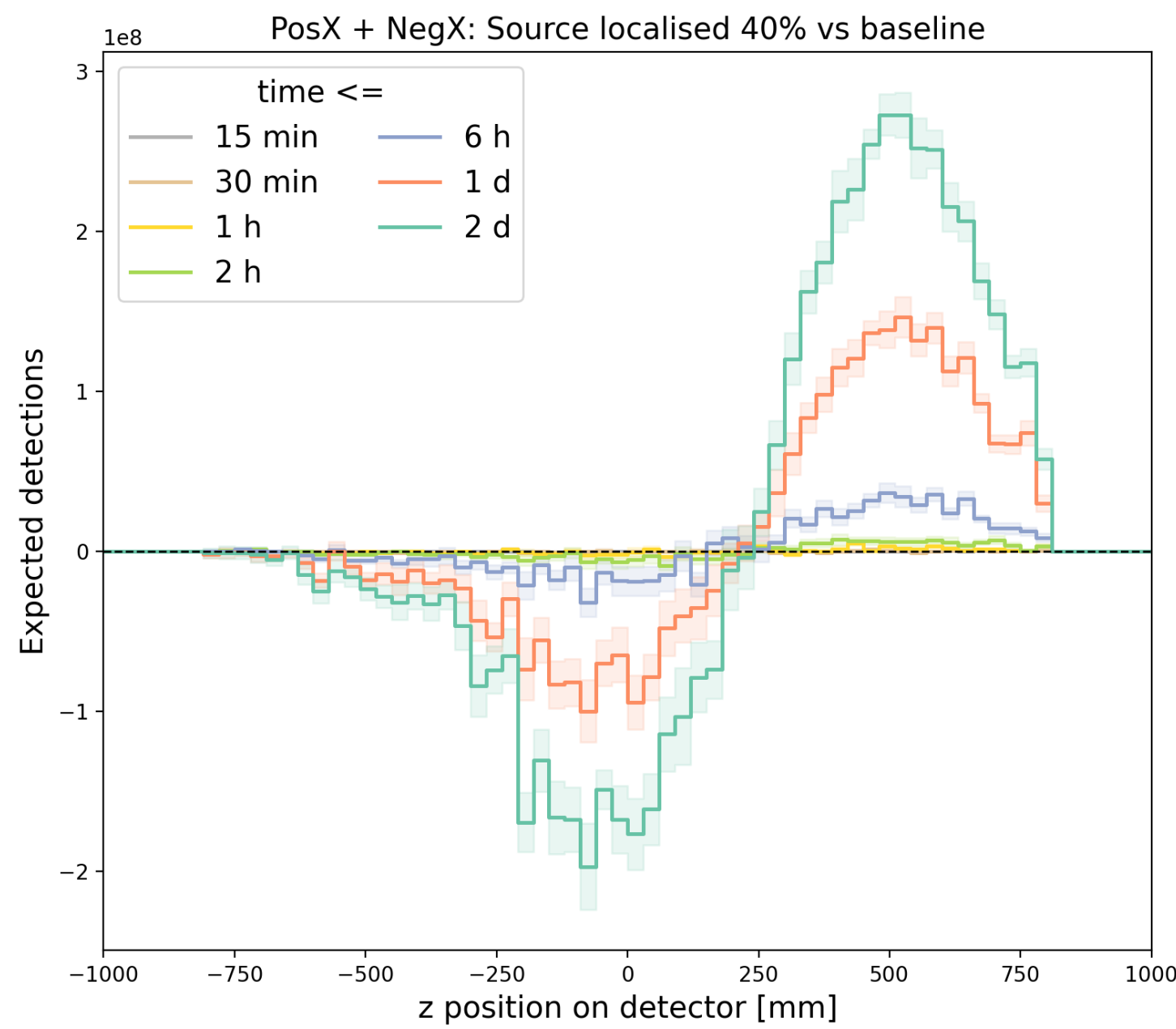
Baseline-subtracted profiles

Detector-position profiles were compared with the retained-source baseline. Baseline subtraction removes the expected glue-retained signal and highlights redistribution-dependent residuals. Broad diffusion gives a wider residual pattern, while localised displacement creates a shifted excess near the displaced source position.



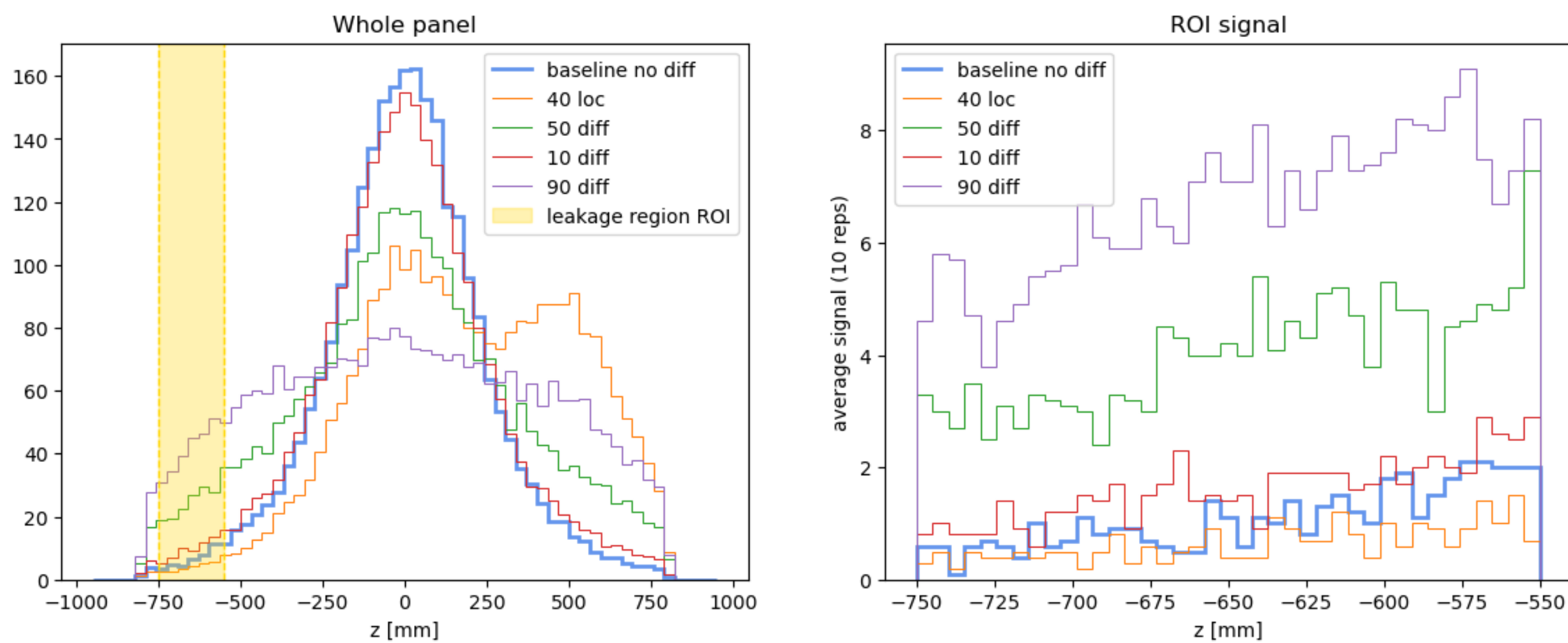
Time-window detectability

Cumulative detector hits were rebinned in increasing time windows and compared with the retained-source baseline. The plotted residuals show how the spatial signal grows as more decay-chain gamma events are collected. A redistribution was counted as detectable when the candidate-baseline difference exceeded the run-to-run baseline variation by at least 3 sigma. Larger redistributions reached this threshold within 30 minutes to 2 hours; smaller signal rearrangements required longer integration times.



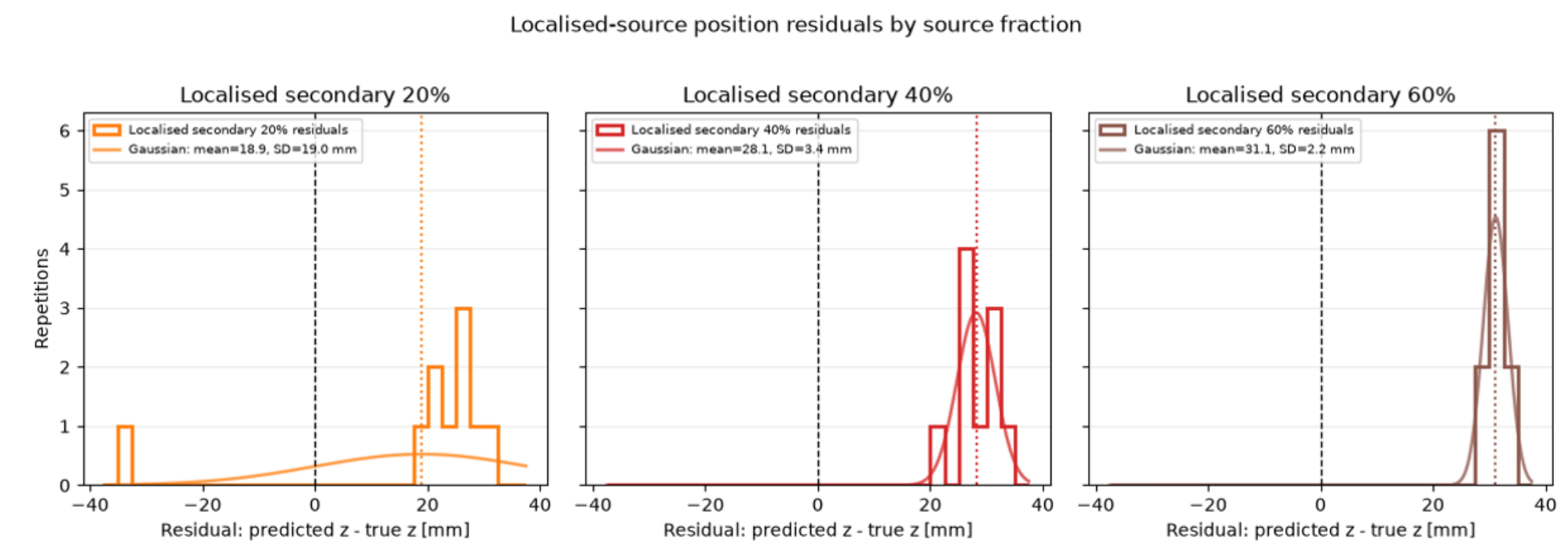
Leakage signal

The leakage analysis tests for excess signal in a displaced-region ROI, away from the retained-source position. Counts in this ROI are compared with the retained-source baseline. A positive excess indicates activity outside the expected glue region, consistent with broad diffusion or localised source accumulation.



Diffusion and localisation detection results

Baseline-subtracted detector profiles were converted into source-monitoring metrics. A regression model trained on profile-shape features recovered broad diffusion fraction with $R^2 \approx 0.998$. Localised displacement was tested separately using a secondary-source search ROI: positive residual signal in this region identified a displaced component, and its centroid estimated source position. Reconstructed positions were approximately 525-531 mm for a simulated 500 mm displaced source.



Conclusions

Gamma detection provides source-localisation information beyond total count rate alone. Although total detector response changed only weakly between retained and redistributed source states, spatial profiles separated retained, broadly diffused, and locally displaced activity. These results support post-treatment monitoring after AlphaGlue delivery and motivate future studies with detector response, background, and patient-scale geometries.

References

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